14.6.1. Hub

It is a common connection point for devices in a network. It joins together the workstations, printers, and servers on a network to communicate with each other. Each hub has a number of ports that connect it to the other devices via a network cable. A hub connects all the devices on its ports together. When data arrives at one port, it is sent to the other ports so that all the devices can see all the information, commonly called **packets**.



Figure 14.17 Hub

Switch

A **switch** is a hardware device that filters and forwards network packets. A network switch also connects computers to each other, like a hub. When a switch receives a packet of data, it determines what computer or device the packet is intended for and sends it to that computer only. It does not broadcast the packet to all computers as a hub. For this reason alone, switches are usually preferred over a hub.



Figure 14.18 Switch

Repeater

Repeaters remove the unwanted noise in an incoming signal. It increases a signal's strength, so it can be transmitted and received over a greater distance without loss in quality. Network repeaters receive and retransmit incoming electrical, wireless or optical signals.

Whenever a repeater receives a signal through one of its ports, it repeats or sends the incoming signal onto the other port. Its main use is to amplify and regenerate signals.